P1 电源树

File: demo_mainboard_p1.kicad_sch

P2 SoC 主控

File: demo_mainboard_p2.kicad_sch

P3 启动存储

File: demo_mainboard_p3.kicad_sch

P4 显示双出口

File: demo_mainboard_p4.kicad_sch

P5 MIPI-CSI 传感器

File: demo_mainboard_p5.kicad_sch

P6 高速差分与连接器

File: demo_mainboard_p6.kicad_sch

P7 板级接口与测试点

File: demo_mainboard_p7.kicad_sch

P8 时钟·复位·调试

File: demo_mainboard_p8.kicad_sch

P9 跨铰链界面

File: demo_mainboard_p9.kicad_sch

P10 充电板(镜框)

File: demo_mainboard_p10.kicad_sch

each sheet = one CSV page P1..P10
10 functional sheets · generated from HW/netlist_v1.csv
self-drawn, not ODM

Sheet: /
File: demo_mainboard.kicad_sch

Title: DEMO MAINBOARD — SCHEMATIC INDEX / 原理图目录

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 1/11

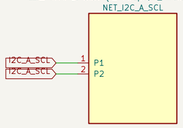
5.0V ±5%
RAIL_VBUS_5V
VBUS_5V
J_TYPE_C
座 (充电板侧, 物理脚位 A/8 编号)
P4 J_TYPE_C
VBUS_5V
GND
USB_CC1
USB_CC2
USB_D_P
USB_D_N
NTC (贴电芯, 另一端接 TS 专用回流)
P18 R_NTC_BAT
ESD 阵列 [待选型]
D_ESD_TC
USB_VBUS_RAW
GND
π 滤波磁珠 [待选型]
FB_VBUS
VBUS_5V_ESD
VBUS_5V_FI
[待选型] 10µF
C_VBUS_IN
[待选型] 100µF
C_VBUS_OUT
[待选型] 100µF
C_VBUS_GND
BQ25896
U_CHG
VBUS_5V_CHG
REGN
SW
PWR_GOOD
CHG_STAT
ILIMIT
NC
SYS_CHG
PGND
单端
NET_BAT_PROT
BAT_CHG
跨铰链 1.5A (双 pin 并联)
RAIL_BAT_CHG
跨铰链 1.5A (双 pin 并联)
J_CHG
HINGE_RSV
HINGE_CHG
TS_BAT
30.31k ±1%
R_TS_PULL
TS_BAT
PGND
4.2V 满 / 3.0V 截止
RAIL_BAT
BAT
BQ2970
U_BATPROT
BAT
BAT_COUT
BAT_DOUT
432243 3.7V ≥420mAh
BT_432243
BAT_CELL_P
BAT_CELL_N
[待选型] N-MOS
Q1
BAT_DOUT
BAT_GATE
[待选型] N-MOS
Q2
BAT_PROT_CELL2
BAT_COUT
3.0–4.2V
RAIL_SYS
SYS
单端
NET_TS_BAT
[待选型] 1µF
C_REGN
REGN
5.24k ±1% (RT1: REGN 到 TS)
R_TS_TOP
TS_BAT
390Ω ±1% (IINMAX ≈1.00A · ICHG ≈0.82A · 单 pin 0.411A)
R_ILIM
IINLIM
VIN (输入)
U_PWR_SW
BAT_SW
PWR_ON
10µF ±20%
C_SW_IN
10µF ±20%
C_SW_OUT
10k ±1%
R_PD_ON
[待选型] **P-MOS 高边开关** (低漏, Vgs(th) 明确)
Q_PG
PWR_ON
BAT_SW
PWR_GOOD
跨铰链段压降测点
TP_BAT_CHG
[待原厂] (典型 0.8–0.9V)
RAIL_V_CORE
V_CORE
5.0V (F13: 由 CARRIER_BOARD_EN 门控)
RAIL_BOARD_5V
SYS_PCB
10µF
C_BULK_V_CORE
100nF
C_HF_V_CORE
[待原厂] (3.3V / 1.8V / 1.2V)
RAIL_V_SOC_IO
V_SOC_IO
10µF
C_BULK_V_SOC_IO
100nF
C_HF_V_SOC_IO
2.8V
RAIL_SENSOR_AVDD
SENSOR_AVDD
GND (pin2)
U_LDO_S28
10µF
C_BULK_SENSOR_AVDD
100nF
C_HF_SENSOR_AVDD
1.8V
RAIL_SENSOR_VDD_IO
SENSOR_VDD_IO
EN (pin3)
U_LDO_S18B
10µF
C_BULK_SENSOR_VDD_IO
100nF
C_HF_SENSOR_VDD_IO
1.2V
RAIL_SENSOR_CORE
SENSOR_CORE
EN (pin3)
U_LDO_S12
10µF
C_BULK_SENSOR_CORE
100nF
C_HF_SENSOR_CORE
3.3V
RAIL_LIGHT_3V3
LIGHT_3V3
EN (pin3)
U_LDO_L33
10µF
C_BULK_LIGHT_3V3
100nF
C_HF_LIGHT_3V3
1.8V
RAIL_LIGHT_1V8
LIGHT_1V8
EN (pin3)
U_LDO_L18
10µF
C_BULK_LIGHT_1V8
100nF
C_HF_LIGHT_1V8
3.3V
RAIL_VDD_3V3_IO
VDD_3V3_IO
EN (pin3)
U_LDO_IO33
10µF
C_BULK_VDD_3V3_IO
100nF
C_HF_VDD_3V3_IO
1.8V
RAIL_VDD_1V8_IO
VDD_1V8_IO
EN (pin3)
U_LDO_IO18
10µF
C_BULK_VDD_1V8_IO
100nF
C_HF_VDD_1V8_IO
ESD 阵列 [待选型]
D_ESD_LIGHT
LIGHT_5V
机壳地 / 数字地 (单点汇流)
RAIL_GND
GND
单点汇流位
CHG_GND_STAR
ESD 阵列 [待选型] (USB 2.0 双通道)
D_ESD_USB_D
USB_D_P_SRC
Type-C 16p 座 (demo 板 5V 输入 + USB 数据)
J_USB_C
RAIL_DEMO_5V
USB_CC1
USB_CC2
ESD 阵列 [待选型] (USB 2.0)
D_ESD_DEMO
π 滤波磁珠 [待选型]
FB_DEMO
10µF ±20%
C_VBUS_DEMO
5.1k ±1% (CC1 下拉)
R_CC1_D
5.1k ±1% (CC2 下拉)
R_CC2_D
22Ω ±1%
R_SER_USB_D_P
NET_USB_D_P_TYPEC
22Ω ±1%
R_SER_USB_D_N
NET_USB_D_N_TYPEC
1M ±1% (上臂)
R_VBUS_DET_D1
1M ±1% (上臂/下臂中点)
NET_VBUS_DET_D
VBUS_DET_D
1M ±1% (下臂)
R_VBUS_DET_D2
100nF ±10%
C_DET_VBUS_D
VBUS_DET_D
1M+1M 分压 + 100nF
J_SOM_P1
VBUS_DET
EN 输出
J_SOM_P3
CARRIER_BOARD_EN
RAIL_DEMO_5V
RAIL_BOARD_5V
EN 输入 (高=放行底板)
U_EN_SW

net = global label · generated from HW/netlist_v1.csv
sheet 2 · 83 devices · paper A2
self-drawn, not ODM

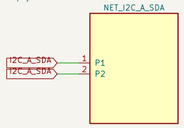
Sheet: /P1 电源树/	
File: demo_mainboard_p1.kicad_sch	
Title: P1 电源树 — DEMO MAINBOARD SCHEMATIC	
Size: A2	Date: 2026-11-11
KiCad E.D.A. 10.0.8	Id: 2/11

Rev: draft-2

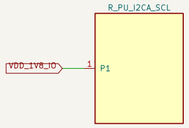
上拉 4.7k 到 3.3V (漏 702µA, t_r=2.38µs@100pF)
NET_I2C_A_SCL



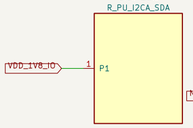
上拉 4.7k 到 3.3V
NET_I2C_A_SDA



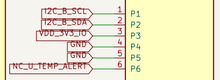
2.2k ±1%
R_PU_I2CA_SCL



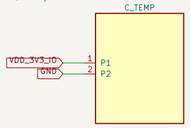
2.2k ±1%
R_PU_I2CA_SDA



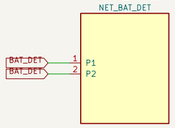
TMP117 数字温度计 (I2C 地址 0x48, ±0.1°C max, 16bit)
U_TEMP



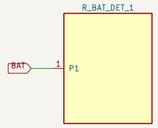
100nF ±10%
C_TEMP



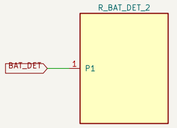
1M+1M 分压 + 100nF
NET_BAT_DET



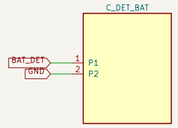
1M ±1%
R_BAT_DET_1



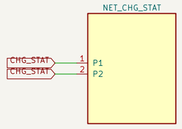
1M ±1%
R_BAT_DET_2



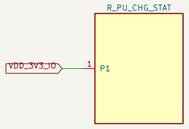
100nF ±10%
C_DET_BAT



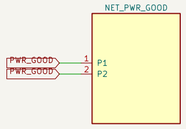
单端
NET_CHG_STAT



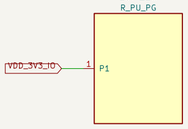
10k ±1%
R_PU_CHG_STAT



单端
NET_PWR_GOOD



10k ±1%
R_PU_PG

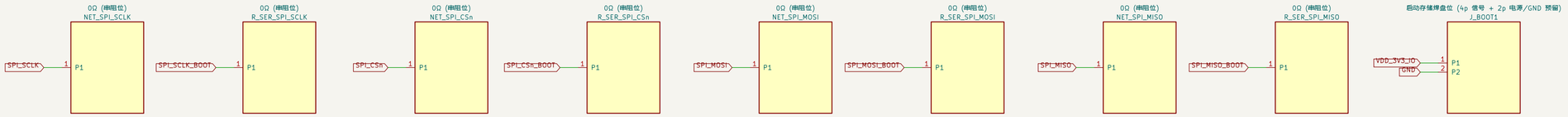


net = global label · generated from HW/netlist_v1.csv
sheet 3 · 14 devices · paper A2
self-drawn, not ODM

Sheet: /P2 SoC 主控/
File: demo_mainboard_p2.kicad_sch

Title: P2 SoC 主控 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 3/11

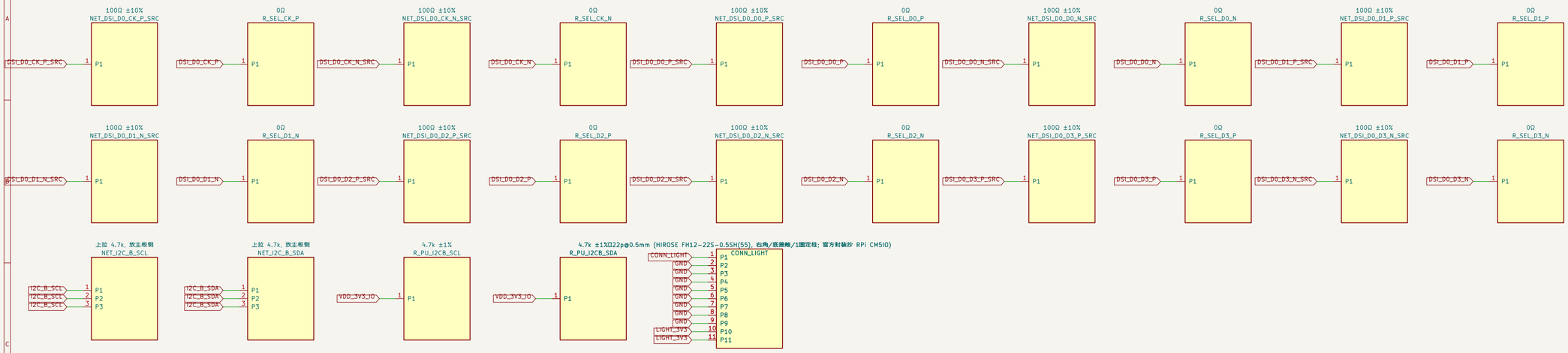


net = global label · generated from HW/netlist_v1.csv
sheet 4 · 9 devices · paper A2
self-drawn, not ODM

Sheet: /P3 启动存储/
File: demo_mainboard_p3.kicad_sch

Title: P3 启动存储 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 4/11

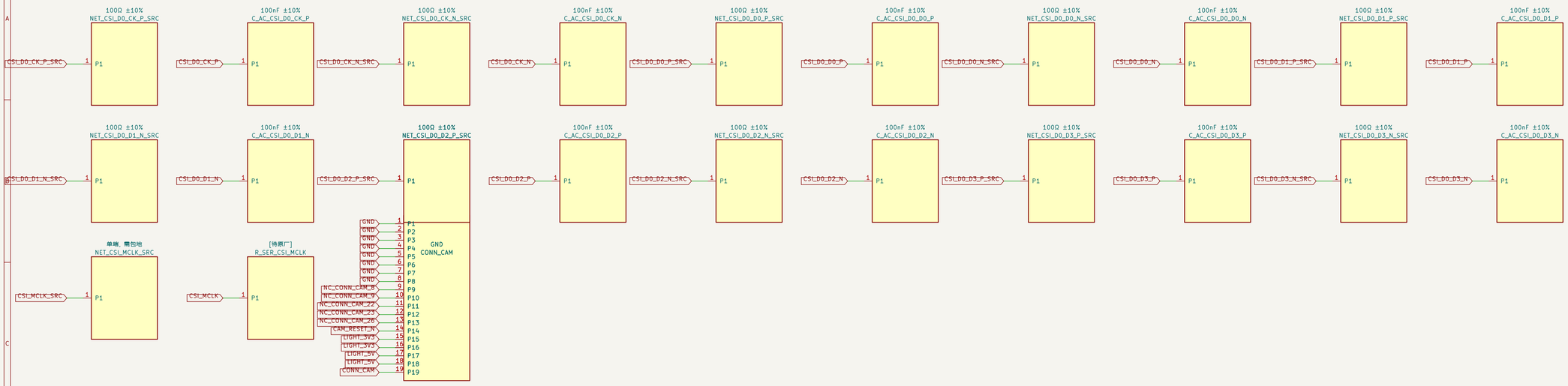


net = global label · generated from HW/netlist_v1.csv
sheet 5 · 25 devices · paper A2
self-drawn, not ODM

Sheet: /P4 显示双出口/
File: demo_mainboard_p4.kicad_sch

Title: P4 显示双出口 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 5/11



net = global label · generated from HW/netlist_v1.csv
sheet 6 · 23 devices · paper A2
self-drawn, not ODM

Sheet: /P5 MIPI-CSI 传感器/
File: demo_mainboard_p5.kicad_sch

Title: P5 MIPI-CSI 传感器 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 6/11



net = global label · generated from HW/netlist_v1.csv
sheet 7 · 6 devices · paper A2
self-drawn, not ODM

Sheet: /P6 高速差分与连接器/
File: demo_mainboard_p6.kicad_sch

Title: P6 高速差分与连接器 — DEMO MAINBOARD SCHEMATIC		
Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 7/11

GPIO 编码 (2 位)
NET_ID_DETECT_0
ID_DETECT_0
P1
0Ω (跳线位)
R_ID_JMP0
ID_DETECT_0
GND
P1
P2
10k ±1%
R_PU_ID0
VDD_3V3_IO
P1
GPIO 编码 (2 位)
NET_ID_DETECT_1
ID_DETECT_1
P1
0Ω (跳线位)
R_ID_JMP1
ID_DETECT_1
GND
P1
P2
10k ±1%
R_PU_ID1
VDD_3V3_IO
P1
1k 串阻 → LED → GND
NET_LED_R
LED_R
P1
1k ±1%
R_LED_R
LED_R_D
P1
LED (特选型)
D_LED_R
GND
P1
1k 串阻 → LED → GND
NET_LED_G
LED_G
P1
1k ±1%
R_LED_G
LED_G_D
P1
LED (特选型)
D_LED_G
GND
P1
1k 串阻 → LED → GND
NET_LED_B
LED_B
P1
1k ±1%
R_LED_B
LED_B_D
P1
LED (特选型)
D_LED_B
GND
P1
上拉 10k，按下到 GND，支持长按
NET_KEY_PWR
KEY_PWR
P1
按键 (特选型)
SW_KEY_PWR
KEY_PWR
GND
P1
P2
10k ±1%
R_KEY_PWR
VDD_3V3_IO
P1
100nF ±10%
C_KEY_PWR
KEY_PWR
P1
同上
NET_KEY_VP
KEY_VOL_P
P1
按键 (特选型)
SW_KEY_VP
KEY_VOL_P
GND
P1
P2
10k ±1%
R_KEY_VP
VDD_3V3_IO
P1
100nF ±10%
C_KEY_VP
KEY_VOL_P
P1
同上
NET_KEY_VM
KEY_VOL_M
P1
按键 (特选型)
SW_KEY_VM
KEY_VOL_M
GND
P1
P2
10k ±1%
R_KEY_VM
VDD_3V3_IO
P1
100nF ±10%
C_KEY_VM
KEY_VOL_M
P1
同上
NET_KEY_CAM
KEY_CAM
P1
按键 (特选型)
SW_KEY_CAM
KEY_CAM
GND
P1
P2
10k ±1%
R_KEY_CAM
VDD_3V3_IO
P1
100nF ±10%
C_KEY_CAM
KEY_CAM
P1
测试点
TP_VBUS
VBUS_5V
P1
测试点
TP_SYS
SYS
P1
测试点
TP_BAT
BAT
P1
测试点
TP_VCORE
V_CORE
P1
测试点
TP_VIO
V_SOC_IO
P1
测试点
TP_LIGHT_3V3
LIGHT_3V3
P1
测试点
TP_CSI_CK_P
CSI_D0_CK_P
P1
测试点
TP_CSI_CK_N
CSI_D0_CK_N
P1
测试点
TP_CSI_D0_P
CSI_D0_D0_P
P1
测试点
TP_CSI_D0_N
CSI_D0_D0_N
P1
测试点
TP_MCLK
CSI_MCLK
P1
测试点
TP_DSI_CK_P
DSI_D0_CK_P
P1
测试点
TP_DSI_CK_N
DSI_D0_CK_N
P1
测试点
TP_DSI_D0_P
DSI_D0_D0_P
P1
测试点
TP_DSI_D0_N
DSI_D0_D0_N
P1
测试点
TP_DSI_D1_P
DSI_D0_D1_P
P1
测试点
TP_DSI_D1_N
DSI_D0_D1_N
P1
测试点
TP_DSI_D2_P
DSI_D0_D2_P
P1
测试点
TP_DSI_D2_N
DSI_D0_D2_N
P1
测试点
TP_DSI_D3_P
DSI_D0_D3_P
P1
测试点
TP_DSI_D3_N
DSI_D0_D3_N
P1
测试点
TP_I2CB_SCL
I2C_B_SCL
P1
测试点
TP_I2CB_SDA
I2C_B_SDA
P1
测试点
TP_I2CA_SCL
I2C_A_SCL
P1
测试点
TP_I2CA_SDA
I2C_A_SDA
P1
测试点
TP_USB_D_P
NET_USB_D_P_TYPEC
P1
测试点
TP_USB_D_N
NET_USB_D_N_TYPEC
P1
测试点
TP_FSIN
FSIN_OUT
P1
测试点
TP_CHG_STAT
CHG_STAT
P1
测试点
TP_PG
PWR_GOOD
P1
测试点
TP_ID
ID_DETECT_0
P1
测试点
TP_LED_R
LED_R
P1
测试点
TP_LED_G
LED_G
P1
测试点
TP_LED_B
LED_B
P1
测试点
TP_KEY_PWR
KEY_PWR
P1
测试点
TP_KEY_VP
KEY_VOL_P
P1
测试点
TP_KEY_VM
KEY_VOL_M
P1
测试点
TP_RESET
RESET_N
P1
测试点
TP_BOOT
BOOT_N
P1
测试点
TP_JTAG
JTAG_TCK
P1
测试点
TP_UART
UART_TX
P1

net = global label · generated from HW/netlist_v1.csv
sheet 8 · 72 devices · paper A2
self-drawn, not ODM

Sheet: /P7 板级接口与测试点/
File: demo_mainboard_p7.kicad_sch

Title: P7 板级接口与测试点 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 8/11

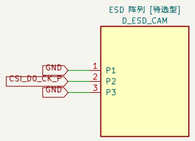
net = global label · generated from HW/netlist_v1.csv
sheet 9 · 40 devices · paper A2
self-drawn, not ODM

Sheet: /PB 时钟·复位·调试/
File: demo_mainboard_PB.kicad_sch

Title: PB 时钟·复位·调试 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 9/11

10k ±1%
R_PG_PULL
PG_AND
VDD_1V8_IO
net
PG 开漏并联 (两路 DC-DC)
NET_PG_AND
[转换型] N-MOS (硬件 AND)
Q_AND
RESET_N
PG_AND
EN 输入 (高 = 使能)
U_DCDC_VCORE
EN_VCORE
V_CORE
[转换型] N-MOS (反相)
Q_ENVC
EN_VCORE
EN_VCORE
10k ±1%
R_EN_VCORE
VDD_1V8_IO
上拉 10k + RC(1k/100nF)
NET_RESET_N
RESET_N
晶振焊盘位 (不装)
X1
XTAL
12pF ±10%
C_XTAL_OUT
XTAL_OUT
GND
12pF
NET_XTAL_OUT
XTAL_OUT
BOOT 按键 (转换型)
SW_BOOT
BOOT_N
GND
单端
NET_UART_TX
UART_TX
单端
NET_UART_RX
UART_RX
1×2 1.27mm 排针 (UART)
J_UART
UART_RX
单端, 50Ω 串阻
NET_FSIN
FSIN
50Ω ±1%
R_SER_FSIN
FSIN_OUT
PC (开漏, 需上拉)
TP_PG_VCORE
PG_AND
P6 (开漏, 需上拉)
TP_PG_VIO
PG_AND
[转换型] 分压上臂 (0Ω 跳线位)
R_SUPV_DIV1
V_SOC_IO
[转换型] 分压下臂 (0Ω 跳线位)
R_SUPV_DIV2
SUPV_SENSE
GND
100nF ±10%
C_SUPV
VDD_1V8_IO
GND
单端, 100Ω 串阻
NET_JTAG_TCK_SOC
JTAG_TCK_SOC
100Ω ±1%
R_JTAG_TCK
JTAG_TCK
单端, 100Ω 串阻
NET_JTAG_TMS_SOC
JTAG_TMS_SOC
100Ω ±1%
R_JTAG_TMS
JTAG_TMS
单端, 100Ω 串阻
NET_JTAG_TDI_SOC
JTAG_TDI_SOC
100Ω ±1%
R_JTAG_TDI
JTAG_TDI
单端, 100Ω 串阻
NET_JTAG_TDO_SOC
JTAG_TDO_SOC
100Ω ±1%
R_JTAG_TDO
JTAG_TDO
1×4 1.27mm 排针 (JTAG)
J_DEBUG
J_DEBUG
复位按键 (转换型)
SW_RST
RESET_N
GND
100nF ±10%
C_RST
RESET_N_SOC
GND
1k ±1%
R_SER_RESET_N
RESET_N_SOC
10k ±1%
R_PU_RESET_N
VDD_1V8_IO
上拉 10k
NET_BOOT_N
BOOT_N
10k ±1%
R_PU_BOOT_N
VDD_1V8_IO
12pF ±10%
C_XTAL_IN
XTAL_IN
GND
12pF
NET_XTAL_IN
XTAL_IN
CT 脚 (断电/延时选择)
U_SUPV
SUPV_CT
VDD_1V8_IO
SUPV_SENSE
RESET_N
EN 输入 (高 = 使能)
U_DCDC_VIO
EN_VIO
V_SOC_IO

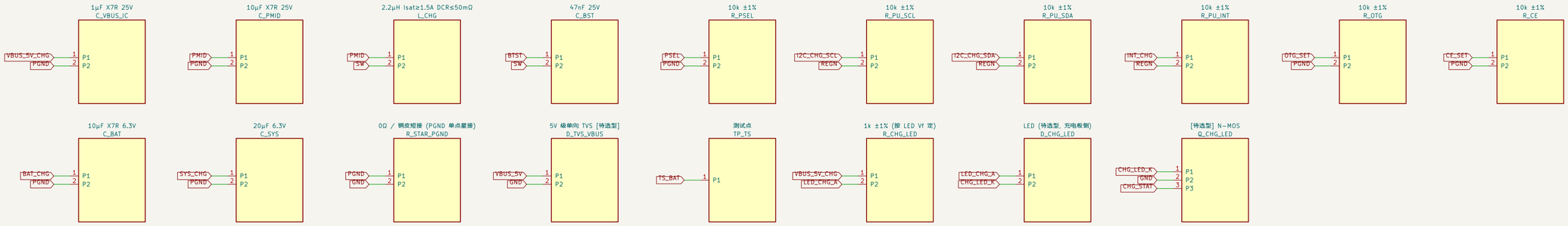


net = global label · generated from HW/netlist_v1.csv
sheet 10 · 1 devices · paper A2
self-drawn, not ODM

Sheet: /P9 跨较链界面/
File: demo_mainboard_p9.kicad_sch

Title: P9 跨较链界面 — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 10/11



net = global label · generated from HW/netlist_v1.csv
sheet 11 · 18 devices · paper A2
self-drawn, not ODM

Sheet: /P10 充电板(独板)/
File: demo_mainboard_p10.kicad_sch

Title: P10 充电板(独板) — DEMO MAINBOARD SCHEMATIC

Size: A2	Date: 2026-11-11	**Rev: draft-2**
KiCad E.D.A. 10.0.8		Id: 11/11